

SQLite, the Tiny Serverless Embedded RDBMS

SQLite is the most popular tiny database engine and has made its way into almost every computing device. Smartphones, either running Android or iOS, are likely to have apps that use it to store data. This article explores SQLite and how it can be used to store your application data.



Although RDBMS' are now four decades old and have matured, things are beginning to change in the realm of database management systems, with the entry of NoSQL and NewSQL data management paradigms. However, enterprise efforts to shift to these new paradigms for their exploding data have been slow, owing to the relative maturity of RDBMSs. At the other end of the spectrum is this tiny SQL db engine that has been dominating its space. SQLite started out as a project to develop an embedded database with a design motive of not requiring any installation or administrator for maintenance, for a United States Navy program. It's now widely used in mobile and Web-based desktop applications. The database engine is as small as 350 KB and supports most of the SQL 92 standard. It was developed by D Richard Hipp, and later the code was released to the public, making it one of the highly active open source projects. Its relatively small size and the wide open source community support have been very appealing for developers, leading them to embed it in their

applications. Although the database has official bindings for C/C++ and TCL only, the community-supported bindings are available for as many as 25 languages.

Distinctive features

The database requires no configuration to set up, and the entire data, as well as any configuration needed, is stored in a single flat file on the disk. This was the main motive behind its development. Also, the database can explode up to the maximum file size on the underlying filesystem.

It is serverless and does not require any process of its own running—and so can be statically linked or embedded in the application itself. It can also be used as an in-memory database.

It supports full text search and is also ACID (atomicity, consistency, isolation, and durability) compliant. It is implemented in C, and works on any platform. It is available as part of the OS in smartphone OSs like iOS and Android.

SQLite has gained a large user base because of its large community and the availability of bindings for all the recent popular languages.